



Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VI Semester – B.E. III & ECE 'B'

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): K.Priya, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit V / Quick Sort
- **Course Outcome:** CO5
- **Topic Learning Outcome/UP:** TLO12
- **Activity Chosen:** Flash Cards
- **Justification:**

A **flash card activity** is a learning exercise that uses flash cards—small cards with questions, terms, or concepts on one side and answers or explanations on the other. It's an effective, interactive way for learners to review and memorize information, as well as reinforce their understanding of a topic.

A flash card activity for Quick Sort is a structured learning exercise designed to help students understand the key concepts, processes, and variations of the Quick Sort algorithm.

- **Time Allotted for the Activity:** 15 Minutes
- **Pre Implementation:**
 - Two types of Flash cards were prepared. One containing question and other card containing answers as shown in Fig 1.
 - Flash card with questions is not shuffled with the cards containing answers
- **Details of the Implementation:**
 - Questions and answers were prepared in a card.
 - Question cards were distributed to certain group of students and answer cards were distributed to few group of students.
 - Student who is having the answer read out the card. Other students listened to the question and found out the question for that answer.
 - Student who is having the respective question card explained the question and answer for the question to the entire class.
 - Questions and Answers for each card are as follows
 - **Card 1:** What is Quick Sort?
Answer: A divide-and-conquer sorting algorithm that sorts elements by selecting a pivot and partitioning the array.



- **Card 2:** Time Complexity of Quick Sort
Answer: Average case: $O(n \log n)$; Worst case: $O(n^2)$.
- **Card 3:** First Step in Quick Sort
Answer: Choose a pivot element.
- **Card 4:** Partitioning in Quick Sort.
Answer: Rearranging elements so that those less than the pivot is on the left and those greater are on the right.
- **Card 5:** What happens after partitioning?
Answer: Recursively apply Quick Sort to the left and right sub-arrays.
- **Card 6:** Methods for Pivot Selection
Answer: First element, last element, median-of-three, or random selection.
- **Card 7:** Impact of Pivot Selection
Answer: A poor pivot can lead to inefficient sorting ($O(n^2)$ in the worst case).
- **Card 8:** When to Use Quick Sort
Answer: Best used for large datasets and when average-case performance is more important than worst-case guarantees.
- **Card 9:** Quick Sort vs. Merge Sort
Answer: Both are $O(n \log n)$, but Quick Sort is in-place, while Merge Sort requires additional space.
- **Card 10:** What is a partition function?
Answer: A function that rearranges the array based on the pivot and returns the pivot's final index.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
CO5	3	3	2	2	3	1	2	1	1

(1 – Low 2 – Moderate 3 – High)



Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
Justification for correlation	Students will be able to apply basic knowledge of engineering in different sorting and searching algorithms.	Students will be able to identify relevant sorting, searching problem for the given applications.	Students will be able to interpret the complex engineering problems with the help of the algorithms.	Students will be able to apply research-based knowledge to identify the suitable sorting and searching algorithms.	Students will be able to use Visualgo.net tool for performing various operations using sorting and searching algorithms.	Students will be able to perform individually the appropriate searching and sorting operations for the given application.	Students will be able to explain and present different searching and sorting algorithms.	Student will become aware of the need for lifelong learning and the continued upgrading of technical knowledge using the concept of searching and sorting techniques.	Students will be able to implement different sorting and searching algorithms in Information and Communication Technology application.

• **Images / Screenshot of the practice:**

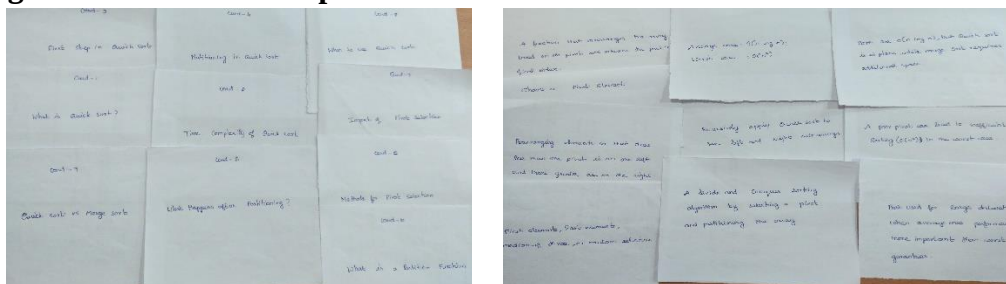


Fig1: Sample Flash Cards



Fig 2: Students (Visalini K and Murugesh AB) involved in Flash Card Activity



□ Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Many students found the flash card activity engaging and appreciated the interactive format. They enjoyed the opportunity to work in groups and quiz each other, which made learning feel more dynamic.
- Students reported that the flash cards helped clarify key concepts of Quick Sort, especially the steps involved and the differences between partitioning techniques.
- Working in pairs or small groups was a highlight for many students. They valued the opportunity to discuss concepts with peers, which facilitated deeper understanding and knowledge sharing.

❖ Benefit of the practice:

- This activity Strengthens memory to the students by actively recalling information.
- It encourages repeated exposure to key concepts, enhancing retention.
- Activity adds a fun, interactive element to learning, which can help with focus and motivation.

❖ Challenges faced in implementation:

- A few students expressed concern that they could recall definitions and processes during the activity but struggled to retain the information long-term.
- While some students were very engaged, others participated minimally, resulting in uneven involvement across groups.
- Some students misinterpreted the prompts on the flash cards, leading to confusion and incorrect answers during discussions

References:

- <https://en.wikipedia.org/wiki/Flashcard> o <https://ellii.com/blog/10-ways-to-use-flashcards-in-class>
- https://www.ritrjpm.ac.in/images/computerscience/13_CS6703_Falshcard
- <https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners>

Signature of Faculty Member

HOD-CSE